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Final Year Project Report

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Abstract

Acknowledgments

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Chapter 1: Introduction

Chapter 2: Literature Review

2.1 Introduction

2.1.1 Clinical Background and the “Black Box” Challenge

In modern oncology, properly identifying and managing brain tumours continues to be a major issue. Due to its simplicity and excellent volumetric resolution, Magnetic Resonance Imaging (MRI) has become the highest standard for analysis (Menze *et al.*, 2015). However, there is a substantial barrier in the interpretation of this data. At the moment, the process of defining the boundaries of tumors is mostly done by hand. Subjectivity and labour intensity are introduced by this reliance on human interpretation, which may not be sustainable for scaling healthcare solutions (Iftikhar *et al.*, 2025). Automated technologies that can match or surpass human precision without the corresponding time restrictions are therefore desperately needed in medical applications.

The response to this clinical need has been the rapid adoption of Deep Learning (DL) technologies, particularly 3D Convolutional Neural Networks (CNNs) like the 3D U-Net. These models have demonstrated “superior performance” in handling volumetric data compared to traditional methods (Bhati, Neha and Amiruzzaman, 2024; Iftikhar *et al.*, 2025). However, the integration of these advanced models has introduced a new, critical issue: the “**black box**” problem.

While these models are highly accurate, their internal decision-making processes are opaque (Neri *et al.*, 2023). **This creates a paradox in medical AI:** as models become more complex and accurate, they often become less interpretable to the clinicians using them. In a high-stakes environment like neurosurgery, a lack of transparency creates a “trust gap” (Wen *et al.*, 2025). It is not enough for a model to simply output a segmentation mask; clinicians require the ability to verify why specific voxels were classified as tumour tissue. Therefore, the lack of interpretability is no longer just a technical issue, but a barrier to ethical and safe clinical deployment (Neri *et al.*, 2023).

2.1.2 Research Aims and Scope of Literature Review

To address this disconnect between model accuracy and clinical trust, this literature review aims to critically evaluate the intersection of 3D deep learning, Explainable AI (XAI), and immersive visualisation. While these fields are often studied in isolation, this review argues that a unified framework is necessary to fully illuminate the “black box” of 3D CNNs.

The review is structured to build this argument logically:

Section 2.2 establishes the technical capabilities of modern 3D CNN architectures for segmentation.

Section 2.3 critically analyses current XAI methods (such as Grad-CAM and LIME), evaluating their limitations when applied to complex volumetric data.

Section 2.4 explores the potential of immersive technologies (VR) to present these XAI outputs in a way that is intuitively understandable for clinicians.

Section 2.5 synthesises these findings to identify the specific research gap this project will address: the lack of an integrated, 3D-visualised explainability pipeline for brain tumour segmentation.

Section 2.6 explains the investigation’s findings and explains how this initiative will fill the identified gap.

2.2 3D Deep Learning for Medical Brain Tumour Segmentation

2.2.1 Evaluation of 3D CNNs for Brain Tumour Segmentation

Over the past decade, the field of medical image analysis has experienced a significant paradigm shift, moving from “**reasoning-based**” systems that relied on manual feature engineering to “**learning-based**” deep learning (DL) approaches (Neri *et al.*, 2023). In the past, radiomics was used for the automated analysis of brain tumours. To train classifiers like Support Vector Machines (SVMs), domain specialists manually retrieved quantitative features including texture, intensity, and shape. However, this method was labour-intensive, prone to observer bias, and frequently failed to capture the intricate, non-linear spatial relationships present in volumetric data (Bhati, Neha and Amiruzzaman, 2024).

The introduction of Convolutional Neural Networks (CNNs) brought an end to this manual paradigm. CNNs prefer an end-to-end approach to extract features automatically by utilizing raw pixel data in contrast to classical machine learning. Initial implementation in neuro-oncology treated Magnetic Resonance Imaging (MRI) as a series of 2D slices. While this approach allowed taking pre-trained networks into consideration (such as VGG or ResNet), it has a significant drawback: it neglects the spatial context along the z-axis (depth).

In neuro-oncology, a tumor is an inherently volumetric entity; it defines the anatomical boundaries with continuation across the sagittal, coronal and axial planes. When a 3D volume is processed as a series of 2D slices it frequently results in “**discontinuous predictions**”, which produce uneven and anatomically impossible 3D models where a lesion is found in one slice but overlooked in the adjacent one (Iftikhar *et al.*, 2025). To overcome this problem, the field has moved towards 3D CNNs, which simultaneously conduct convolutions on the entire (x, y, z) volume using volumetric kernels (such as $3 \times 3 \times 3$). This volumetric approach allows the model to learn complex spatial hierarchies and inter-slice connections, which is essential for precise segmentation of brain tumors using large-scale datasets like BraTS.

2.2.2 Evolution of Volumetric Segmentation Architectures

To handle the computational complexity of 3D data while conducting segmentation that is pixel-perfect, the research community has standardized the “**encoder-decoder**” architecture.

Building upon the success of 2D U-Net, Çiçek et al. (Çiçek *et al.*, 2016) introduced the 3D U-Net architecture, a fully convolutional network for dense volumetric image segmentation. A U-shaped symmetric model consists of two distinct pathways:

1. **The Contracting Path (Encoder):** This pathway significantly considers feature extraction. It consists of max-pooling layers after recursive blocks of 3D convolutions. As data is processed through the encoder, the spatial resolution decreases while the number of features increases. This path allows the model to capture high-level conceptual data context (“where is the tumour present”), such as differentiating tumour tissues from healthy ones (Çiçek *et al.*, 2016).
2. **The Expanding Path (Decoder):** To regenerate the segmentation map, the spatial resolution that was lost during encoding needs to be restored. This path reconstructs the feature maps back to their initial input dimensions using up-convolution (transposed convolution) layers, also providing precise localisation (“where is the tumour”).

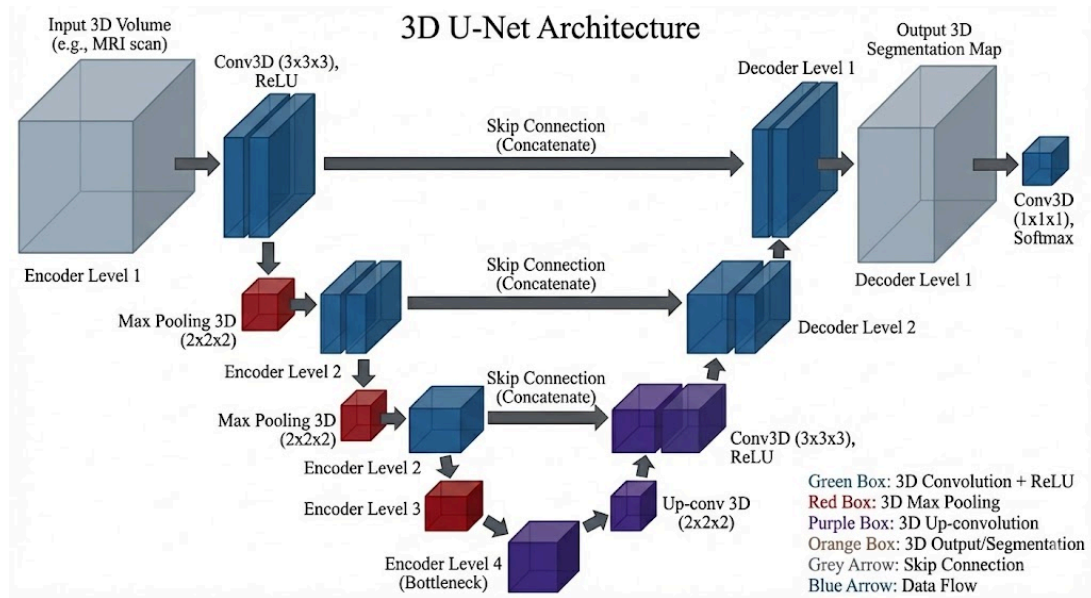


Figure 1: 3D U-Net Architecture

A critical innovation of the 3D U-Net introduced long-range skip connections. Deep neural networks often face difficulty acquiring fine-grained spatial information after applying multiple pooling operations. The 3D U-Net addresses this problem by concatenating high-resolution feature maps straight from the encoder to the appropriate decoder layers. According to Çiçek et al. (Çiçek *et al.*, 2016), this mechanism allows the network to incorporate deep semantic features with shallow details, with precise boundary delineation even when trained with sparse annotations.

Despite its success, the standard U-Net struggles with small lesions. Medical images often contain irrelevant background information such as non-considerable tissue in the brain, while the standard U-Net treats all the pixels in a feature map equally. To overcome this issue, researchers have integrated “attention mechanisms” into the U-Net architecture. As explained

by Natekar et al. (Natekar, Kori and Krishnamurthi, 2020), the Attention U-Net incorporates “**attention gates**” into the skip connections.

These gates utilise the coarser signal from the gating vector (the deep layer) to filter the activations from the input signal (the shallow layer) before they are merged. This enhances the activations in the Region of Interest (ROI) and suppresses the irrelevant background regions using mathematical equations. This “soft-attention” mechanism improves the model’s sensitivity to small, irregular lesions such as glioma subregions without requiring additional guidance or complex cropping preprocessing (Natekar, Kori and Krishnamurthi, 2020).

Parallel to the U-Net, Milletari et al. (Milletari, Navab and Ahmadi, 2016) proposed the V-Net, an optimised model specifically for volumetric medical data. V-Net is completely distinct from other models, incorporating **residual connections** (short-skip connections) blocks. This model resembles the ResNet philosophy, adding the input of a block to its output, allowing gradients to flow more smoothly during backpropagation. Compared to convolutional U-Net, this reduces the vanishing gradient issue and makes it easier to train much deeper 3D networks (Milletari, Navab and Ahmadi, 2016).

Furthermore, the V-Net research has presented a solution to one of the most enduring challenges in medical segmentation: class imbalance. In a dataset, tumour occupies less than 1% of the total volumetric pixels while the rest of the area is unaffected and healthy background. Standard objective functions like cross-entropy fail in this scenario, resulting in a model achieving accuracy of 99% by predicting “background irrelevant data” for every voxel. To overcome this, Milletari et al. (Milletari, Navab and Ahmadi, 2016) introduced the **Dice coefficient**, which optimises the model by considering the gap between the ground truth and the expected segmentation:

$$D = \frac{2 \sum_i^N p_i g_i}{\sum_i^N p_i + \sum_i^N g_i}$$

Where the sums run through the N voxels, of the predicted binary volume $p_i \in P$ and the ground truth binary volume $g_i \in G$. This function can be improved by rewriting it in a differentiable form. As a result, it can be optimised with gradient descent. It can be computed with respect to each predicted voxel, allowing the network to improve the overlap between the predictions and ground truth (Milletari, Navab and Ahmadi, 2016).

The gradient:

$$\frac{\partial D}{\partial p_j} = 2 \left[\frac{g_j}{\sum_i p_i^2 + \sum_i g_i^2} - \frac{2p_j(\sum_i p_i g_i)}{(\sum_i p_i^2 + \sum_i g_i^2)^2} \right]$$

2.3 Explainable AI (XAI) for Explainability in Medicine

2.3.1 The Need for Explainability in Clinical Practice

The integration of Artificial Intelligence (AI) into medical practice, especially in high-stakes domains like neuro-oncology, has been made more difficult by the opacity of modern Deep Learning (DL) models. Convolutional Neural Networks have demonstrated superior performance in segmentation-like tasks, yet their complex, non-linear decision-making remains inaccessible. This “black box” nature has become a barrier to model adoption, often referred to as the “**trust gap**” (Wen *et al.*, 2025).

Just model prediction is insufficient in clinical practice; it must incorporate justification that follows medical knowledge. As highlighted by Neri *et al.* (Neri *et al.*, 2023), the ethical and legal framework governing medicine, such as the “right to explanation” under GDPR, demands that automated decisions be transparent and contestable. Clinicians must be able to identify the reason behind the model classification of a specific region as tumour-present or not to ensure patient safety (Iftikhar *et al.*, 2025).

Furthermore, XAI is not just a compliance tool but also a critical mechanism for model debugging and knowledge discovery. Researchers can detect “clever Hans” events, in which a model learns misleading connections (e.g., forecasting tumour based on a specific hospital watermark rather than anatomical pathology) (Hou *et al.*, 2024). When it comes to brain tumour segmentation, XAI refers to the validation of network focus on biologically relevant sub-regions such as the necrotic core or enhancing tumour (Natekar, Kori and Krishnamurthi, 2020).

2.3.2 Review of Voxel-Based XAI Techniques

The literature categorizes XAI approaches according to their scope and timing relative to model training. A primary distinction is made between ante-hoc (or “intrinsic”) and post-hoc methods.

- **Ante-hoc:** Methods designed to be interpretable by their nature. For example, decision trees or linear regression models, where the relationships between them are explicit (Agrawal *et al.*, 2025). By integrating interpretable prototype blocks directly into the network architecture, recent advancements in “Self-Explainable AI (S-XAI)” plan to introduce transparency in deep learning (Hou *et al.*, 2024).
- **Post-hoc:** Methods designed for explainability after the model has been trained. They treat a model as a “black box” and try to improve and approximate its behaviour based on the given inputs and outputs. This method is preferred as standard for analysing medical imaging models like the 3D U-Net architecture to perform segmentation (Bhati, Neha and Amiruzzaman, 2024).

Further XAI methods are divided by their scope:

- **Global explanations:** Methods that explain the overall logic of the model across the entire dataset, such as feature importance in a model.

- **Local explanations:** Methods that explain the decision-making process of the model for specific inputs, such as the reason behind classifying a voxel region as tumour. Local explanation is important for clinical decision-making as they offer the patient-specific rationale for diagnosis and therapy planning (Iftikhar *et al.*, 2025).

For 3D medical imaging, the most common XAI techniques are those that can generate saliency maps, spatial representations of “feature importance”. The three most prominent approaches are **Grad-CAM**, **LIME** and **SHAP**.

Gradient-Weighted Class Activation Mapping (Grad-CAM) has been recognised as the “golden standard” for visualising CNN decisions in medical imaging (Bhati, Neha and Amiruz-zaman, 2024). Grad-CAM is model-agnostic and can be applied to any CNN model without retraining, compared to earlier methods that required architectural changes.

Mechanism: Grad-CAM utilises the gradient of the target concept (e.g., tumour) to produce a saliency map, which highlights the regions of the input image that contribute most to the prediction of the target class. It computes a weighted sum of the feature maps, where the weights represent the importance of each feature channel to the prediction. To eliminate the negative contributions, a Rectified Linear Unit (ReLU) is then applied, highlighting more specific regions that positively support the class of interest (Selvaraju *et al.*, 2017).

Application in 3D: Natekar et al. (Natekar, Kori and Krishnamurthi, 2020) successfully applied Grad-CAM to a 3D U-Net for brain tumour segmentation, as it was originally designed for 2D images. Their research showed that 3D Grad-CAM can localise tumours based on the aggregated gradients across volumetric feature maps. However, it has a critical limitation which is the resolution trade-off: because Grad-CAM uses feature maps from deep layers, the resulting heatmaps are often coarse and may bleed into surrounding tissue, reducing the voxel-level precision of the segmentation mask itself (Natekar, Kori and Krishnamurthi, 2020).

Local Interpretable Model-Agnostic Explanations (LIME) takes a different approach. LIME treats the model as a function only, without using internal gradients.

Mechanism: LIME modifies the model’s predictions and observes the change by masking out random superpixels (segments). It then fits a simple, interpretable model to these disrupted samples to predict the model’s behaviour locally around a specific image (Ribeiro, Singh and Guestrin, 2016).

Evaluation in 3D: While LIME is highly recommended for 2D image segmentation, its performance in 3D is computationally expensive. According to Agrawal et al. (Agrawal *et al.*, 2025), LIME requires thousands of forward passes to generate stable explanations. For data like 3D images which contain millions of voxels, generating valid superpixels and running sufficient distributions to achieve significant statistics is often too slow for real-time clinical workflows. Additionally, defining the “superpixels” in a 3D brain volume is quite difficult, frequently leading to unstable explanations that vary between runs.

SHAP (SHapley Additive exPlanations) provides a theoretically sound substitute based on cooperative game theory. Each feature (voxel) is given an “importance value” that represents its marginal contributions to the prediction, averaged over all possible combinations of features (Ribeiro, Singh and Guestrin, 2016).

Strengths and Weaknesses: SHAP provides the most mathematically consistent explanations, by conforming the summation of each feature’s attributes to be equal to the model’s output. However, it has extreme computational costs in high-dimensional spaces. “DeepSHAP,” an approximation method, has been used to accelerate this process, but research suggests that SHAP values can mislead the explanation in deep networks, as they may violate axioms when features are highly correlated, which is often common in statistically coherent MRI data (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

2.3.3 Comparative Synthesis and the Fidelity Gap

All three methods have their strengths and weaknesses, but Grad-CAM remains the best performing method for 3D medical image segmentation because of its computational efficiency (only one backward pass is needed) and its capacity to make use of the spatial information present in CNN feature maps (Natekar, Kori and Krishnamurthi, 2020).

However, a critical “fidelity gap” exists, as existing measures frequently show significant discrepancy between the predicted and ground truth segmentation (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025). This discrepancy is exacerbated by the standard practice of analysing 3D attention maps as static 2D slices — a method that hides volumetric coherence and increases cognitive load for clinicians. Therefore, closing the “trust gap” calls for a paradigm shift towards immersive virtual reality, which provides the high-dimensional, intuitive visualisation needed to properly understand these deep volumetric insights, rather than simply algorithmic improvements.

2.4 Immersive Visualisation in Medical Imaging

The interpretation of volumetric medical data has historically been analysed on 2D screens, which causes a lot of cognitive stress. According to Khedir et al. (Khedir *et al.*, 2025), traditional slice-by-slice navigation requires clinicians to mentally reconstruct volumetric architecture. This process requires time and is prone to spatial errors when assessing the depth of the tumours relative to prominent brain regions. Even though 3D Convolutional Neural Networks (CNNs) can process this volumetric data, the representation of their outputs remains largely stuck in 2D.

Emerging research suggests that immersive visualisation such as Virtual Reality (VR) and Augmented Reality (AR) can bridge this gap. VR provides **stereoscopic depth perception** and **6-Degrees-of-Freedom (6DoF)** interaction which goes well beyond standard monitors. This allows users to view the “saliency maps” not as flat overlays, which are generated by XAI, but as volumetric clouds suspended in 3D space. The NeuroXAI framework has shown that immersive environments significantly improve clinicians’ ability to localise pathology and understand complex neural connectivity compared to standard desktop viewers (Zeineldin *et al.*, 2022).

The application of VR extends well beyond passive viewing to active surgical planning. Defining tumour boundaries in neuro-oncology is critical. Recent advancements in Augmented Reality (AR) enable surgeons to simulate trajectories and visualise “risk maps” created by deep learning models before entering the operation theatre (Zeineldin, 2023; Khedir *et al.*, 2025). For situations like complex glioma, where 2D scans may not represent the tumour with key blood arteries, this tool is a crucial resection corridor.

Additionally, the idea of interactive AI is essential to current medical systems. Static heatmaps (like standard Grad-CAM) offer “**take it or leave it**” explanations. VR environments, on the other hand, make **Human-in-the-Loop (HITL)** interaction easier. Immersive interfaces enable clinicians to interrogate the model, e.g., by digitally “erasing” a portion of the input volume to identify prediction changes (Orsmaa *et al.*, 2025). This interactivity transforms XAI from a static report into a dynamic conversation between the clinician and the AI, allowing the correction of model errors in real time (Orsmaa *et al.*, 2025).

To implement these immersive systems requires robust technological pipelines. Standard medical formats (DICOM/NIfTI) are not supported by native game engines like Unity. However, platforms like 3D Slicer have filled this bridge. The SlicerVR extension allows volume rendering of medical scans in VR headsets without any data conversion loss. Despite this, integrating XAI with VR still continues to be a major difficulty. Most pipelines concentrate on 2D slice-based visualisation, which fails to convey the volumetric feature importance and limits the clinical utility of XAI in more in-depth scenarios (Zeineldin *et al.*, 2022). There is a need for distinct workflows that can consume a 3D CNN’s attention weights and render them as interactive volumetric objects, as proposed in frameworks like DeepIGN (Zeineldin, 2023).

2.5 Synthesis and Research Gap

2.5.1 The Disconnect Between Model Accuracy and Clinical Utility

After addressing and reviewing the three most significant chapters of this review, a distinct contrast has been established. Models like the 3D U-Net and its variants have achieved tremendous success in segmenting brain tumours (Çiçek *et al.*, 2016; Milletari, Navab and Ahmadi, 2016). On the other side, the opacity of the “black box” remains a critical barrier to clinical trust (Neri *et al.*, 2023). Although methods like Grad-CAM have successfully illuminated the internal logic of these networks (Natekar, Kori and Krishnamurthi, 2020), current implementations suffer from three major limitations that this project aims to address.

2.5.2 Identified Gaps in Current XAI Frameworks

Having global achievements in 3D XAI based on research such as NeuroXAI, there is still something which is called an “**interactivity gap**”; current systems use static visualisation rather than active user contribution (Zeineldin *et al.*, 2022). This passivity leads to a serious methodological error that confuses feature relevance with model confidence. As mentioned in the DeepIGN study (Zeineldin, 2023), a network may exhibit high focal attention on a specific region while displaying high epistemic uncertainty. This could give a confident result but incorrect false positives. As a result, existing 3D XAI methods do not provide the required mechanisms for **interactive correction or uncertainty quantification**, leaving a functional gap where the user cannot distinguish between a confident prediction and a statistical guess, or change the model’s focus (Abyasa and Rahmania, 2025).

Furthermore, the utility of these volumetric insights is severely constrained by the “cognitive friction” of traditional visualisation mediums. Khedir *et al.* (Khedir *et al.*, 2025) mentioned that analysing 3D volumetric pathology via slice-by-slice 2D navigation increases significant burden to mentally reconstruct spatial relationships from disconnected images. The mismatch between the 3D nature of the data and the 2D nature of the display limits the interpretability of the feature maps, as crucial depth signals are lost in translation. Therefore, a clear research gap exists for a unique framework that not only generates uncertainty-aware explanations but also represents them in an immersive virtual environment, to clarify whether or not engagement significantly lowers cognitive load compared to conventional medical workflows.

2.6 Conclusion

This literature review establishes that while 3D deep learning architectures, particularly the 3D U-Net, have achieved superior success in volumetric tumour segmentation, their deployment is critically hindered by the “black box” paradox (Neri *et al.*, 2023). Clinical workflows still remain detached even if explainable methods like Grad-CAM provide mathematical solutions for transparency. The reliance on static 2D slice-by-slice visualisation fails to convey the complex spatial resolution of 3D pathologies and is hindered by a significant cognitive load that limits clinical trust (Natekar, Kori and Krishnamurthi, 2020).

Consequently, this research focusing on bridging the “trust gap” requires a paradigm shift from passive observation to an interactive environment. By developing a virtual reality environment that incorporates uncertainty quantification with real-time feedback, this project aims to “illuminate the black box” effectively. This innovative method will ensure that high-performance AI is not just accurate but also understandable and useful for medical practice.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the comprehensive methodology employed in developing an Explainable AI (XAI) framework for brain tumor segmentation using 3D Convolutional Neural Networks (CNNs). The primary objective of this research is to transform an opaque deep learning model into a transparent system that not only achieves high segmentation accuracy but also provides clinically meaningful insights into the decision-making process. This work addresses the critical challenge of removing the “black box” problem in medical imaging AI by implementing six complementary XAI techniques and evaluating them against ground truth annotations.

The research approach follows a hybrid Agile/CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which provides the structural rigor required for medical applications while allowing for iterative model improvements. The framework processes entire brain MRI volumes using SegResNet, a 3D residual encoder-decoder architecture, to produce volumetric tumor predictions across three clinically relevant regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

This chapter is organized as follows: Section 2.2 describes the overall approach and development methodology; Section 2.3 details the implementation including data preprocessing, model architecture, and XAI methods; Section 2.4 presents the research approach and evaluation strategy; Section 2.5 discusses challenges and limitations; and Section 2.6 addresses ethical considerations.

3.2 Approach

3.2.1 Overall Approach and Rationale

This research employs secondary data analysis using the publicly available BraTS 2023 dataset, which comprises approximately 1,250 patient cases with multi-modal MRI scans. The high-level strategy encompasses three interconnected components: (1) training a 3D CNN (SegResNet) to perform volumetric tumor segmentation, (2) integrating multiple XAI techniques to reduce the black-box nature of the model, and (3) exporting segmentation and XAI outputs into a VR-compatible pipeline for immersive clinical evaluation.

The decision to use 3D volumetric processing rather than 2D slice-by-slice analysis is grounded in clinical necessity. Brain tumors are inherently three-dimensional structures, and processing individual slices independently sacrifices critical spatial context that exists between adjacent slices (Milletari, Navab and Ahmadi, 2016; Çiçek *et al.*, 2016). Volumetric approaches preserve this contextual information, leading to more clinically realistic segmentations that better capture tumor boundaries and spatial extent.

SegResNet was selected as the primary architecture due to its residual connections, which provide robustness against vanishing gradients and enable effective training of deep networks

(He *et al.*, 2016). The architecture offers an optimal balance between performance and computational complexity, making it suitable for medical imaging applications where training data may be limited and model interpretability is essential.

The rationale for implementing multiple XAI methods stems from their complementary strengths. Gradient-based methods (Grad-CAM, Guided Backpropagation, LRP) provide different perspectives on feature importance, while perturbation-based approaches (Occlusion Sensitivity) offer model-agnostic explanations. Monte Carlo Dropout adds uncertainty quantification, enabling identification of regions where the model lacks confidence—a critical capability for clinical decision support.

3.2.2 Agile CRISP-DM Methodology and Its Use

Traditional waterfall development lifecycles are ill-suited for the unpredictable, highly experimental nature of deep learning model training. The complexity of neural network architectures, combined with the need for extensive hyperparameter tuning and iterative refinement, demands a more flexible approach. This research adopts a hybrid Agile/CRISP-DM methodology that combines the structured phases of the Cross-Industry Standard Process for Data Mining with the rapid iteration and evolutionary prototyping principles of Agile development.

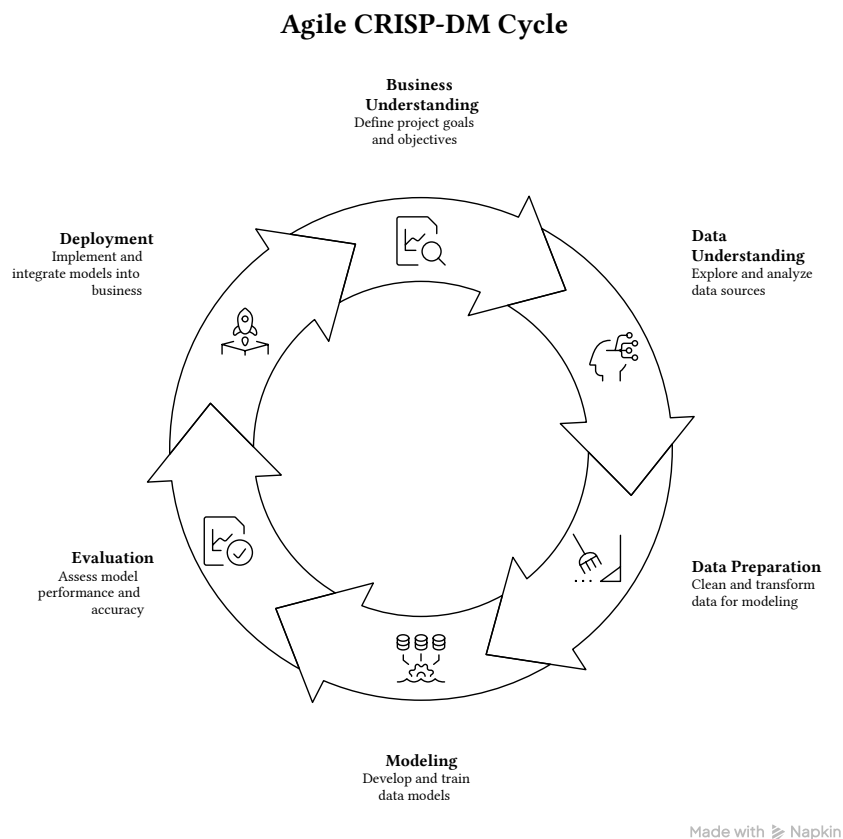


Figure 2: The Agile CRISP-DM methodology employed in this research, showing the iterative cycles between the six core phases.

Figure 2 illustrates the six interconnected phases of the methodology, with dashed arrows indicating the iterative nature of the development process. Each phase feeds into the next, but the Agile approach allows for rapid cycling back to earlier phases when issues are discovered or improvements are identified.

Business and Clinical Understanding. The clinical motivation for this research stems from two fundamental challenges in neuro-oncology: the black-box nature of 3D CNNs limits clinical trust in AI-assisted diagnoses, and traditional 2D slice-by-slice assessment introduces significant cognitive load for radiologists. The project scope was defined as developing a highly accurate 3D automated segmentation framework integrated with a comprehensive suite of Explainable AI methods, with outputs natively supporting immersive Virtual Reality tools for clinical review.

Data Understanding and Preparation. This phase involved extensive exploration of the BraTS 2023 dataset structure, including the four MRI modalities (T1, T1-contrast, T2, FLAIR) and their clinical significance for tumor characterization. A heavily engineered data pipeline was established to handle label conversion (separating WT, TC, and ET regions), foreground cropping to remove empty space around the brain, and intensity normalization to standardize input distributions.

Modeling through Iterative Development. An evolutionary prototyping Agile strategy was employed, beginning with a monolithic prototype trained on a small subset of 45 patients. This prototype served to validate the 3D data pipeline, test candidate model architectures (SwinUNETR, AttentionUnet, SegResNet), and verify hardware memory constraints before committing to full-scale training. After successful prototyping, the codebase was refactored into a scalable PyTorch/MONAI framework, with configuration-driven exploration leading to SegResNet selection for the primary training loop.

Evaluation through Continuous Verification. Model checkpoints were iteratively refined based on strict validation criteria encompassing Dice similarity, Hausdorff Distance 95th percentile (HD95), Intersection over Union (IoU), Sensitivity, and Specificity. XAI methods were integrated one sprint at a time, with targeted quality metrics (Saliency IoU, Uncertainty Area Ratio, Pointing Game) verifying feature relevance and uncertainty characteristics. Detailed metric definitions and quantitative results are presented in subsequent chapters.

Deployment at Integration Level. The final phase generates predictive outputs, heatmaps, and variance arrays as standardized NIfTI files, feeding directly into an immersive 3D Slicer/VR visualization pipeline for clinical review. It is important to note that this remains an experimental research prototype and has not been deployed for active clinical diagnostics.

The Agile CRISP-DM approach ensured an organized, rapid-iteration development loop while maintaining the stringent robustness required for safety-critical medical machine learning components. Regular sprint reviews allowed for course correction, and the modular architecture enabled independent validation of each component.

3.3 Implementation

3.3.1 Data Acquisition and Preprocessing

The BraTS 2023 dataset, provided through the MICCAI Brain Tumor Segmentation Challenge, serves as the foundation for this research. The dataset comprises approximately 1,250 patient cases, each containing four co-registered MRI modalities: T1-weighted (T1), T1-weighted with Gadolinium contrast enhancement (T1c), T2-weighted (T2), and T2 Fluid-Attenuated Inversion Recovery (FLAIR). All volumes have been resampled to an isotropic resolution of 1 mm³ and skull-stripped.

The tumor labels in BraTS follow a standard annotation scheme where each voxel is classified as background (0), necrotic tumor core (1), peritumoral edema (2), or enhancing tumor (3). For this research, these labels are converted into three binary channels representing clinically relevant regions: Whole Tumor (WT = necrosis + edema + enhancing), Tumor Core (TC = necrosis + enhancing), and Enhancing Tumor (ET = enhancing only).

The preprocessing pipeline implements several critical transformations to prepare the data for model training:

- **Foreground Cropping:** Removes empty space surrounding the brain, reducing volume dimensions from approximately 240³ to around 160³ voxels per dimension, significantly reducing computational requirements.
- **Channel-First Formatting:** Reshapes volumes to [4, 160, 160, 160] to conform to PyTorch conventions.
- **Per-Channel Z-Score Normalization:** Computes mean and standard deviation only on non-zero voxels to preserve background structure, then normalizes each modality independently.
- **Random Spatial Augmentation:** During training only, applies random flips (across all three axes), 90-degree rotations, intensity scaling, intensity shifting, and Gaussian noise to improve model generalization.

3.3.2 SegResNet Model Architecture

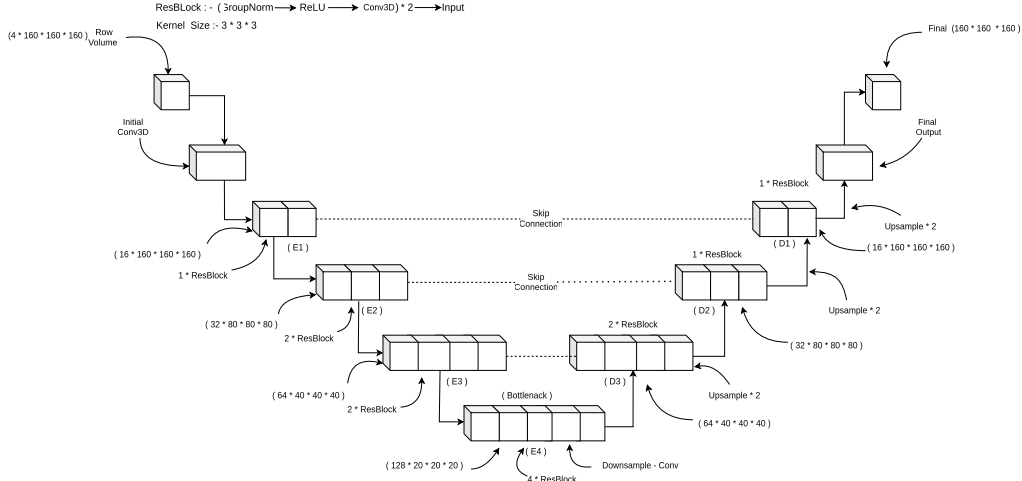


Figure 3: The SegResNet architecture, showing the encoder-decoder structure with residual connections and skip connections.

SegResNet is a 3D encoder-decoder architecture specifically designed for volumetric medical image segmentation. The model processes entire brain MRI volumes without sliding windows during training, enabling global context modeling that captures long-range spatial dependencies.

The encoder progressively downsamples the input volume across four levels using strided convolutions (stride = 2), extracting increasingly abstract features. The filter progression follows: 4 input channels (MRI modalities) $\rightarrow$ 32 filters (Level 1) $\rightarrow$ 64 filters (Level 2) $\rightarrow$ 128 filters (Level 3) $\rightarrow$ 256 filters (Level 4, bottleneck). Corresponding spatial resolutions are $160^3 \rightarrow 160^3 \rightarrow 80^3 \rightarrow 40^3 \rightarrow 20^3$.

At the bottleneck (Level 4), the 256-channel feature maps at 20^3 resolution capture the entire brain tumor context in a highly compressed form. This layer serves as the target for Grad-CAM analysis, providing a coarse but globally-aware representation of tumor location.

The decoder mirrors the encoder but uses trilinear upsampling to restore spatial resolution. Skip connections directly concatenate features between matching encoder-decoder levels, preserving fine-grained spatial information and improving gradient flow during backpropagation. The final output produces 160^3 volumes with 3 channels corresponding to WT, TC, and ET predictions.

Every layer in SegResNet is built from residual blocks (ResBlocks) that learn residual mappings rather than direct transformations. Each ResBlock consists of two 3D convolutions (3^3 kernels) with GroupNorm and ReLU activations, plus a skip connection that adds the input to the output. This architecture provides several benefits: direct gradient paths that solve vanishing gradients, interpretable “correction” learning rather than opaque transformations, and training stability for very deep networks.

The model configuration uses 32 initial filters, a dropout probability of 0.1 (which serves dual purposes of regularization and enabling MC Dropout), and DiceFocalLoss as the training objective. The focal component (with $\gamma = 2.0$) is critical because Enhancing Tumor comprises only approximately 5% of voxels, making it easily overlooked by standard loss functions. Focal loss upweights misclassifications on ET, effectively balancing class imbalance.

3.3.3 Evolutionary Prototyping Phase

Before committing to full-scale training, an evolutionary prototyping phase was conducted using a 45-patient subset randomly sampled from the BraTS training data. This phase served multiple purposes: validating the complex 3D data pipeline, rapidly testing candidate architectures, and verifying hardware memory limits.

The initial prototype was implemented as a monolithic script that combined data loading, model definition, training, and evaluation in a single file. While this approach lacks modularity, it enabled rapid experimentation and quick iteration on architectural choices. After validating core functionality, the prototype was refactored into a modular, configuration-driven PyTorch framework with clear separation between data handling, model definitions, training logic, and evaluation.

Prototype results demonstrated that SegResNet provided the optimal performance-to-complexity ratio among the tested architectures. The 45-patient subset achieved Dice scores of 0.74 (WT), 0.43 (TC), and 0.27 (ET) on a 5-patient test set—promising results given the limited training data and confirming the viability of the approach for full-scale implementation.

3.3.4 Implementation Pipeline

[Space reserved for detailed implementation pipeline description]

The implementation follows a modular architecture designed for reproducibility and extensibility. The codebase is organized into several key components:

- **Configuration Management:** YAML-based configuration files control all hyperparameters, paths, and experiment settings, enabling reproducible experiments and easy parameter sweeps.
- **Data Pipeline:** The `src/data` module handles dataset scanning, patient splitting, and MONAI transform pipelines for both training and inference.
- **Model Factory:** The `src/models` module provides a unified interface for creating models, loss functions, optimizers, and learning rate schedulers from configuration.
- **Training Framework:** The `src/training` module implements the full training lifecycle with automatic mixed precision, best-model checkpointing, crash recovery, and Weights & Biases logging.
- **XAI Suite:** The `src/xai` module contains implementations for all six explanation techniques, with a consistent interface for generating saliency maps and uncertainty estimates.

- **Evaluation Tools:** The src/evaluation module computes both traditional segmentation metrics and novel XAI-specific metrics, exporting results to CSV files for analysis.

The end-to-end workflow proceeds through seven stages: (1) configuration loading and data initialization, (2) model training with validation, (3) test set inference, (4) XAI map generation for all methods, (5) metric computation and CSV export, (6) NIfTI volume export for 3D Slicer, and (7) VR visualization preparation.

3.3.5 Full-Scale Training Setup

The final training configuration was optimized for a single high-memory GPU (RTX 5880 Ada, 48 GB VRAM) with Automatic Mixed Precision (AMP) enabled to maximize training efficiency. Key hyperparameters include:

Parameter	Value
Batch Size	1 (limited by 3D volume memory)
Epochs	35
Learning Rate	5×10^{-5}
LR Scheduler	Cosine Annealing
Optimizer	AdamW (weight decay 1×10^{-5})
Loss Function	DiceFocalLoss
Focal Gamma	2.0
Mixed Precision	Enabled

Table 1: SegResNet training hyperparameters for BraTS 2023 full dataset.

The dataset was split into 1,000 training cases, 125 validation cases, and approximately 125 test cases. The validation set was used for hyperparameter tuning and early stopping, while the test set was reserved for final evaluation and XAI analysis. All splits were performed deterministically using a fixed random seed (42) to ensure reproducibility.

Training progress was monitored using Weights & Biases (W&B) for experiment tracking, with metrics logged at every iteration. Best model checkpoints were selected based on validation Dice score, and the final model was evaluated on the held-out test set to report unbiased performance estimates.

3.3.6 XAI Module Implementation

The XAI framework implements six complementary explanation techniques, each providing unique insights into model decision-making:

Grad-CAM (Gradient-weighted Class Activation Mapping) computes per-feature-map importance weights using gradients flowing into the bottleneck layer. For target class c and feature map k , importance weights are calculated as:

$$\alpha_k^c = \left(\frac{1}{Z}\right) \sum_d \sum_h \sum_w \frac{\partial y^c}{\partial A_{d,h,w}^k}$$

where y^c is the spatially-averaged output logit and Z is the number of spatial elements. The heatmap is computed as $L^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right)$ and upsampled to input resolution. Grad-CAM provides class-discriminative but coarse localization ($20^3 \rightarrow 160^3$).

Guided Backpropagation modifies the standard backpropagation algorithm by adding an additional gating mechanism at every ReLU layer during the backward pass. The modified gradient flow zeros out gradients where either the forward activation or the incoming gradient is negative, keeping only purely positive signal paths. This produces full-resolution (160^3) saliency maps with sharp edges and fine detail, though it is not class-discriminative.

Guided Grad-CAM combines the strengths of both methods through element-wise multiplication: $\text{Guided_Grad-CAM} = \text{GBP} \circ \text{upsample}(\text{Grad-CAM})$. The Grad-CAM heatmap acts as a soft spatial mask, restricting the sharp GBP detail to class-relevant regions. This produces high-resolution, class-discriminative explanations that capture both spatial localization and fine-grained feature attribution.

Layer-wise Relevance Propagation (LRP) propagates the output prediction score backward through the network, distributing relevance based on activation magnitudes. This research uses the $\text{Input} \times \text{Gradient proxy}$, which is mathematically equivalent to ε -LRP for ReLU networks: $R_i = x_i \times \frac{\partial f(x)}{\partial x_i}$. This provides ultra-high resolution relevance maps identifying which input voxels contributed most to the final prediction.

Occlusion Sensitivity tests model reliance on specific regions by systematically occluding (masking) spatial patches and recording the resulting drop in prediction confidence. A sliding 16^3 window moves across the input with stride 8, and at each position, the masked region is set to baseline (0). The score drop $S_{c(i,j,k)} = f_{c(x)} - f_{c(x_{\text{occluded}})}$ indicates region importance. This model-agnostic approach requires approximately 1,000 forward passes per patient but provides robust, gradient-free explanations.

Monte Carlo Dropout quantifies model uncertainty by performing multiple stochastic forward passes with dropout layers active at inference time. For N passes, the mean prediction $\bar{p} = \left(\frac{1}{N}\right) \sum_{n=1}^N p_n$ provides a stabilized output, while the variance $\sigma^2(d, h, w) = \left(\frac{1}{N}\right) \sum_{n=1}^N \left(p_{n(d,h,w)} - \bar{p}(d, h, w)\right)^2$ serves as an uncertainty estimate. High variance indicates regions where the model is inconsistent and therefore less trustworthy.

All XAI outputs are saved as NIfTI volumes organized per patient and per method, ready for loading into 3D Slicer or VR visualization tools. The implementation ensures spatial alignment between saliency maps and ground truth labels by computing metrics inline during generation, using the same preprocessed coordinate space.

3.4 Research Approach

3.4.1 Research Design

This study employs applied, quantitative research using secondary medical imaging data. The experimental design evaluates one primary architecture (SegResNet) combined with six XAI techniques, measuring both segmentation performance and explanation validity against ground truth annotations.

The experimental unit is a single BraTS patient case. Independent variables include the choice of XAI method and the tumor sub-region being evaluated (WT, TC, ET). Dependent variables encompass segmentation metrics (Dice, HD95, IoU, Sensitivity, Specificity), XAI metrics (Pointing Game, Coverage, IoU, Weighted Dice), and uncertainty statistics (UAR, boundary ratios).

3.4.2 Experimental Setup

The computational environment consists of a high-performance workstation with an NVIDIA RTX 5880 Ada GPU (48 GB VRAM), 128 GB system RAM, and an AMD Ryzen Threadripper processor. The software stack includes Python 3.10, PyTorch 2.0+, MONAI 1.3+, and supporting libraries for medical image I/O and visualization.

Reproducibility measures include fixed random seeds where possible, comprehensive version control using Git, and saving configuration files alongside model checkpoints. All experiments are logged to Weights & Biases with full hyperparameter tracking and artifact storage.

3.4.3 Evaluation Strategy

Model performance is evaluated using five standard medical imaging metrics computed on the held-out test set:

- **Dice Score:** Measures volumetric overlap between prediction and ground truth, ranging from 0 (no overlap) to 1 (perfect overlap). This is the primary BraTS benchmark metric.
- **Hausdorff Distance 95th Percentile (HD95):** Measures the worst-case surface boundary error in millimeters, providing sensitivity to boundary accuracy that Dice may miss.
- **Intersection over Union (IoU):** A stricter overlap metric that penalizes false positives more heavily than Dice.
- **Sensitivity (Recall):** The true positive rate, measuring the proportion of actual tumor voxels correctly identified. High sensitivity is critical for clinical safety to avoid missing tumors.
- **Specificity:** The true negative rate, measuring the proportion of healthy tissue correctly excluded. High specificity prevents over-segmentation and false alarms.

XAI quality is evaluated using four metrics that compare saliency maps to ground truth:

- **Pointing Game:** Binary metric indicating whether the voxel with maximum saliency falls inside the tumor region.

- **Saliency Coverage:** The fraction of total saliency mass that falls inside the tumor, measuring attention concentration.
- **Saliency IoU:** Shape overlap between thresholded saliency and ground truth, assessing spatial alignment.
- **Weighted Dice:** A novel metric treating continuous saliency as soft prediction weights, capturing gradual alignment better than binary thresholding.

Uncertainty evaluation includes the Uncertainty Area Ratio (fraction of uncertainty inside tumor), Boundary Uncertainty Ratio (fraction at tumor edges), and correlation between LRP relevance and uncertainty to identify “brittle” decision regions.

3.4.4 Link to Research Questions

This methodological design directly addresses the research questions posed in Chapter 1:

- **Segmentation Accuracy** is measured by traditional metrics (Dice, HD95), answering whether the model achieves clinically acceptable performance.
- **Model Attention Alignment** is assessed by XAI metrics (Pointing Game, Coverage), answering whether the model “looks at the right place for the right reasons.”
- **Reliability Characteristics** are quantified by uncertainty metrics, answering where the model is confident versus uncertain.

By combining these evaluation dimensions, this research can argue not only that the model is accurate, but also that its decision-making process is clinically interpretable and trustworthy.

3.5 Challenges and Limitations

3.5.1 Technical Challenges

Several significant technical challenges were encountered during this research. Memory constraints of 3D CNNs necessitated careful ROI size selection (160^3) and batch size limitation (1), impacting training efficiency. Long training times (approximately 24 hours for 35 epochs) and even longer XAI generation times (especially for Occlusion Sensitivity, requiring 1,000 forward passes per patient) required workflow optimization and selective processing.

Integrating multiple XAI methods into a single coherent 3D framework presented architectural challenges, particularly regarding hook isolation for Guided Grad-CAM. When Grad-CAM and GBP hooks coexist on the same model, they corrupt each other’s gradients, necessitating sequential computation per patient.

Workarounds included using Automatic Mixed Precision to reduce memory usage, limiting XAI runs to representative test subsets, and adjusting occlusion parameters (window size, stride) to balance resolution against computational cost.

3.5.2 Methodological Limitations

This research has several methodological limitations that should be considered when interpreting results. The use of a single benchmark dataset (BraTS 2023) without external clinical

validation limits generalizability to other scanners, populations, or tumor types. Potential domain shift may affect performance if the model is applied to data from different institutions or acquisition protocols.

XAI methods themselves have known shortcomings: Grad-CAM produces blurry heatmaps due to upsampling, GBP is not class-discriminative, and Occlusion Sensitivity is computationally expensive. The XAI metrics serve as proxies for clinical interpretability but may not fully capture how radiologists actually perceive and use explanation maps.

The VR visualization component remains primarily a visualization target rather than an extensively evaluated clinical tool. User studies with practicing radiologists would be necessary to validate the clinical utility of the immersive explanation interface.

3.5.3 Threats to Validity

Internal validity may be threatened by preprocessing settings, train/test split randomness, and hyperparameter choices. External validity is limited by the single-dataset design and lack of cross-institutional validation. Construct validity concerns the extent to which metrics like Pointing Game and Weighted Dice truly measure “correct reasoning” by the model—these are operationalizations of interpretability that may not fully capture the clinical concept.

3.6 Ethical Considerations

3.6.1 Data Privacy and Governance

This research uses only de-identified BraTS data that has been stripped of all personally identifiable information. Data handling practices include local secure storage with restricted access, no attempt at re-identification, and adherence to dataset usage terms. The BraTS challenge operates under appropriate ethical approvals for secondary use of medical imaging data in research.

3.6.2 Clinical Safety and Responsible Use

It is essential to emphasize that this system is a research prototype and has not been approved for clinical decision-making. The intended use is as a decision-support and research tool to explore model behavior, not as an autonomous diagnostic system. Risks include over-reliance on AI outputs and potential misinterpretation of saliency maps or uncertainty maps by non-expert users.

Clear documentation accompanies all outputs explaining what XAI maps represent and what they do not. The system is designed to augment, not replace, clinical expertise.

3.6.3 Bias and Generalisation

The BraTS dataset, while large and diverse, may contain biases related to geography, scanner types, and pathology distribution. Limited external validation means performance may change on under-represented populations. Future work should include broader, more diverse datasets to ensure equitable performance across demographic groups.

3.6.4 Ethical Use of VR for Medical Imaging

Immersive visualization may amplify visual cues (e.g., bright saliency hotspots) and affect clinician perception. Safeguards for future deployment include clear documentation on XAI map interpretation, training materials for clinicians, and emphasis on combining AI insights with expert judgment rather than relying solely on automated outputs.

3.7 Summary

This chapter has presented a comprehensive, modular methodology for generating and evaluating explainability in 3D brain tumor segmentation models. The key contributions include: (1) implementation of six XAI methods adapted for 3D volumetric data, (2) development of four quantitative XAI evaluation metrics including a novel Weighted Dice metric, (3) integration of MC Dropout for uncertainty quantification, (4) a complete evaluation pipeline bridging model performance with explanation validity, and (5) VR-ready outputs for immersive clinical evaluation.

The Agile CRISP-DM methodology provided an organized framework for iterative development while maintaining the rigor required for medical applications. The implementation is modular and reproducible, controlled primarily via YAML configuration files and scripted entry points. The following chapter presents the quantitative results of applying this framework to the BraTS 2023 test set, demonstrating which XAI methods most effectively explain SegResNet decisions and how well they align with ground truth tumors.

Chapter 4: Results

Chapter 5: Evaluation

Chapter 6: Conclusion

Chapter 7: Code

Use code blocks for snippets. Do not dump 1000s of lines here; use appendices for that. [cite: 20, 21]

```
java      static      public      void      main(String[]      args)      {      try
{ UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName()); } catch(Ex-
ception e) { e.printStackTrace(); } new WelcomeApp(); }
```

Listing 1: Java Example

Chapter 8: Evaluation

Include an evaluation of both product and process. Be objective. Reflect on what worked and what did not.

Chapter 9: Conclusion

Reflect on key findings, limitations, and future opportunities.

Chapter 10: Initial Project Proposal

Chapter 11: Final Project Proposal

Chapter 12: Application for Approval of Research Activities

Chapter 13: Client Consent Form

Chapter 13: Bibliography

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